

Chap - 1

Electric circuit $\longrightarrow$ Interconnection of electrical elements.

Charge (C) $\longrightarrow$ Electrical property of the atomic particles of

$1e = 6.24 \times 10^{18}$ elect. which matter consists.

Electric current (A) $\longrightarrow$ time rate of change of change

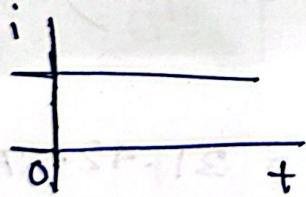
$$i = \frac{dq}{dt}$$

$$1A = 1Cs^{-1}$$

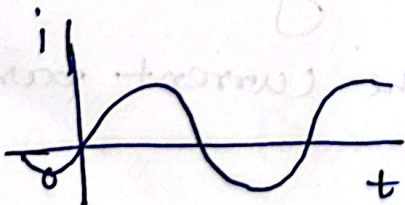
$$Q = \int_{t_0}^t i dt$$

$\rightarrow$ Int eq.

Direct current (DC) $\longrightarrow$ current remains constant with time.

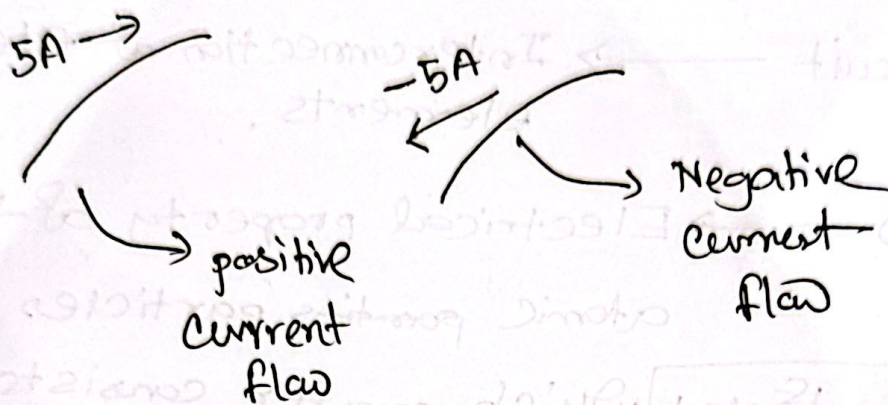


Alternating current (AC) $\longrightarrow$ current varies with time.



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$$1 \text{ electron} = -1.602 \times 10^{-19} \text{ C}$$



Q → The total charge entering a terminal is given by $q = 5t \sin 4\pi t$ mC. Cal. current at $t = 0.5$ s

Ans:
$$i = \frac{dq}{dt} = \frac{d}{dt} (5t \sin 4\pi t) \text{ mA}$$

$$= (5 \sin 4\pi t + 20\pi t \cos 4\pi t) \text{ mA}$$

At $t = 0.5$ s

$$i = 5 \sin 2\pi + 10\pi \cos 2\pi = 31.42 \text{ mA} \quad (\text{Ans})$$

Q → Determine the total charge entering a terminal between $t = 1$ s and $t = 2$ s if the current passing the terminal is $i = (3t^2 - 1) \text{ A}$

$$Q = \int_{t=1}^2 i dt = \int_1^2 (3t^2 - 1) dt = \left[t^3 - \frac{t}{2} \right]_1^2$$

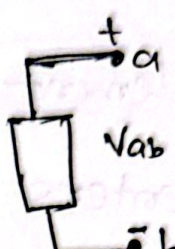
$$= (8 - 2) - (1 - 1/2) = 5.5 \text{ C} \quad (\text{Ans})$$

Voltage / potential difference $(V) \rightarrow$ Energy required to move a unit charge through an element.

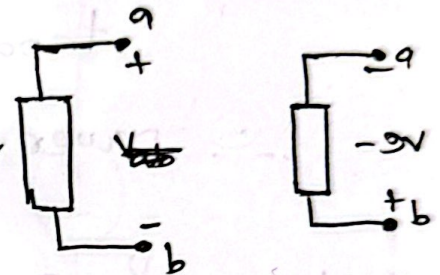
$$V_{ab} = \frac{dw}{dq}$$

$\rightarrow$ Change of work/Energy

$\rightarrow$ Change of charge



$$1V = 1 J C^{-1} = 1 Nm C^{-1}$$



$$V_{ab} = -V_{ba}$$

power $\rightarrow$ time rate of expending or absorbing energy (W)

$$P = \frac{dw}{dt} \rightarrow \frac{dw}{dq} \times \frac{dq}{dt} = Vi$$

$$P = +vi \text{ or } vi > 0 \rightarrow \text{Absorbing power}$$

$$P = -vi \text{ or } vi < 0 \rightarrow \text{Releasing or supplying power.}$$

* Current flow $\rightarrow (+) \text{ to } (-)$

$$E = \int P dt$$

Passive sign convention $\rightarrow$ is satisfied when the current enters through the positive terminal of an element and $p = +vi$. If the current enters through the negative terminal, $p = -vi$

+ power absorbed = - power supplied

$\therefore$ power supplied + power absorbed = 0

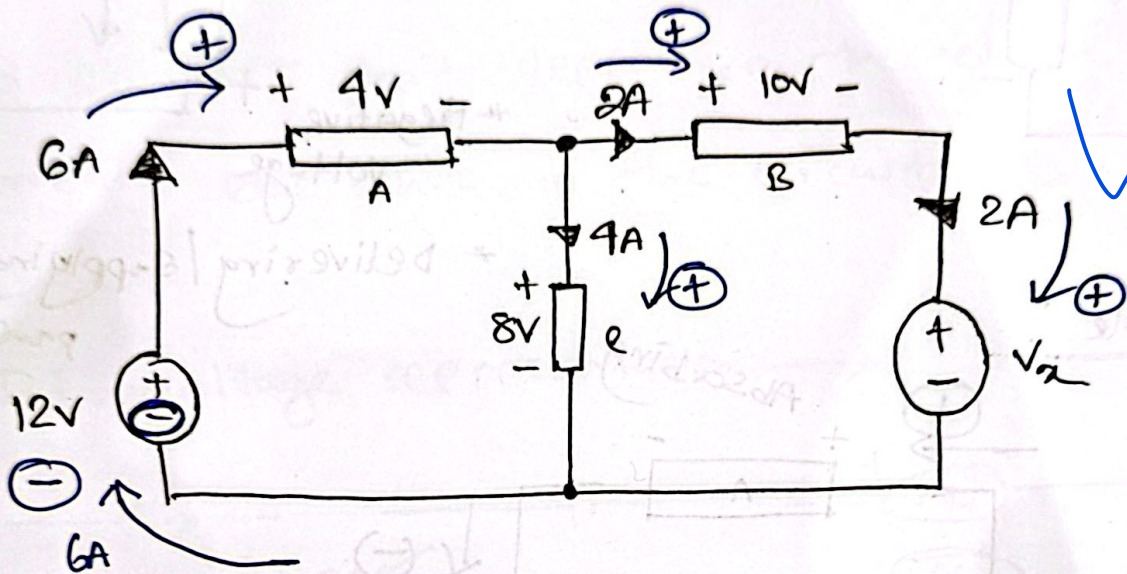
* Law of Energy conservation $\rightarrow \boxed{\sum p = 0}$

$$w = \int_{t_0}^t p \, dt = \int_{t_0}^t v_i \, dt$$

* Energy is the capacity to do work (J)

$$1 \text{ Wh} = 3600 \text{ J}$$

Q → Using passive sign convention and the conservation of Energy, what is the value of voltage V_x ?
 (-) → supplied
 (+) → Absorbed



$$P_A = 6 \times 4 = 24 \text{ W}$$

$$P_B = 2 \times 10 = 20 \text{ W}$$

$$P_C = 4 \times 8 = 32 \text{ W}$$

$$P_{\text{supply}_1} = (+6) \times 12 = -72 \text{ W}$$

minus 9 टक्के

$$P_{\text{supply}_2} = 2 \times V_x = 2V_x$$

∴ Law of conservation of energy,

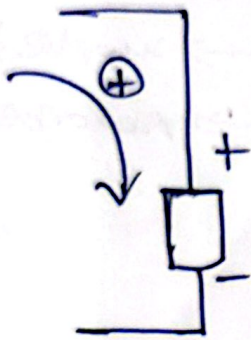
power supplied + power absorbed = 0

$$\Rightarrow -72 + 24 + 20 + 32 + 2V_x = 0$$

$$\Rightarrow 4 + 2V_x = 0$$

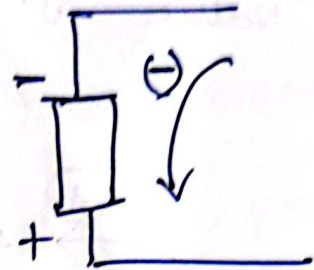
$$\Rightarrow V_x = -2 \text{ V (Ans.)}$$

passive sign convention:



* positive voltage

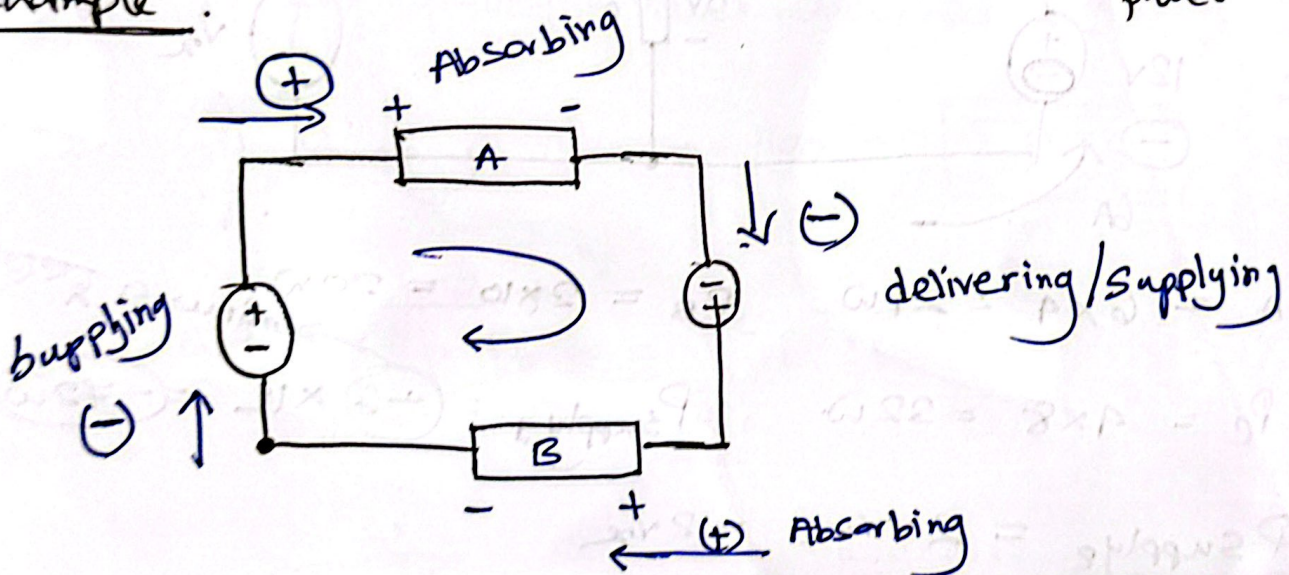
* Absorbing power



* Negative voltage

* Delivering / supplying power

Example:

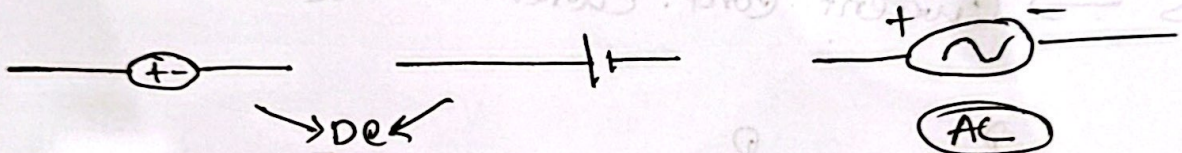


Dependent and Independent Sources

* Independent Source:

The element for which both voltage and current don't depend on the voltage or current elsewhere in the circuit.

Ind. voltage representation →



Ind. Current representation →

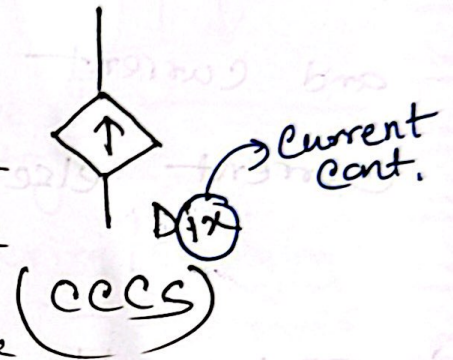
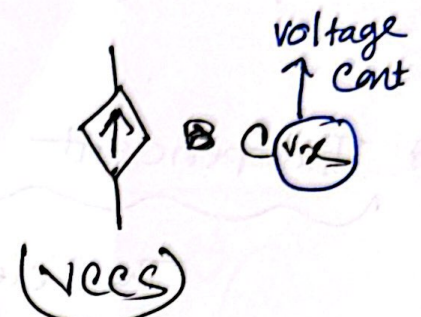
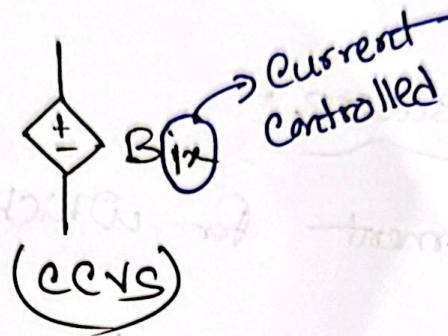
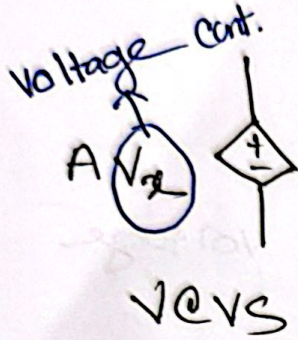


* Dependent Source:-

The element for which either the voltage or current depends on the voltage or current elsewhere in the circuit.

Dep. Voltage Rep. $\rightarrow$

Dep. Current Rep $\rightarrow$

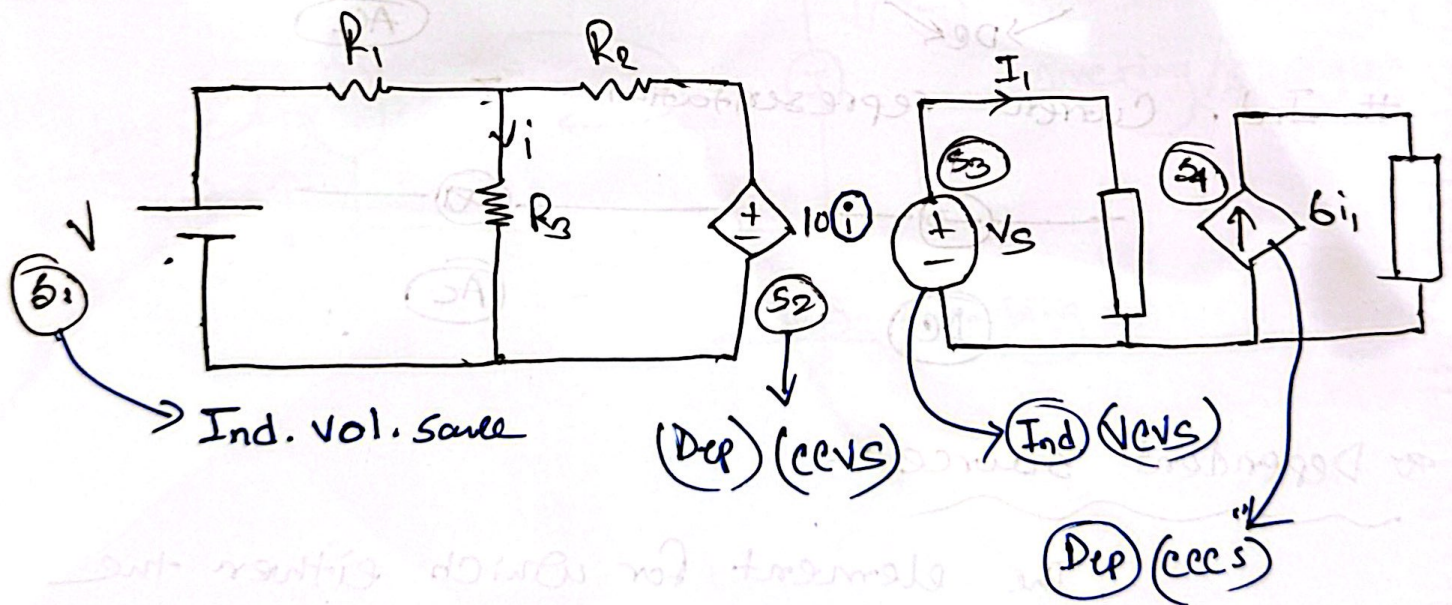


VCVS $\rightarrow$ Voltage cont. Voltage source

CCVS $\rightarrow$ Current cont. Voltage source

VCCS $\rightarrow$ Voltage cont. Current source

CCCS $\rightarrow$ Current cont. Current source



Chap - 2

Ohm's Law:

Ohm's law states that the voltage (V) across a resistor is directly proportional to the current (i) flowing through the resistor.

* Temp. Const

$$V \propto i$$

$$V = IR$$

$$I \propto V$$

* Length and cross-sectional area should be same of the conductor

$$\Rightarrow I = GV$$

$$\Rightarrow I = \frac{1}{R} V$$

$$\Rightarrow V = IR$$

Resistance:- The resistance (R) of an element denotes its ability to resist the flow of electric current.

$$R = \rho \frac{L}{A}$$

Resistivity of material ($\Omega \cdot m$)
Length
cross sectional area

Hinders the flow of electron

* Change in material/temp $\rightarrow$ Change in the value of (R)

$$R \propto L \longrightarrow R \uparrow L \uparrow / R \downarrow L \downarrow$$

$$R \propto \frac{1}{A} \longrightarrow R \uparrow A \downarrow / R \downarrow A \uparrow$$

$$R = \frac{V}{I}$$

$$1 \Omega = 1 \text{ V A}^{-1}$$

* A resistor always absorbs power from the circuits

Semi-conductor

- Carbon
- Germanium
- Silicon

Insulator

- Paper
- Mica
- Glass
- Teflon

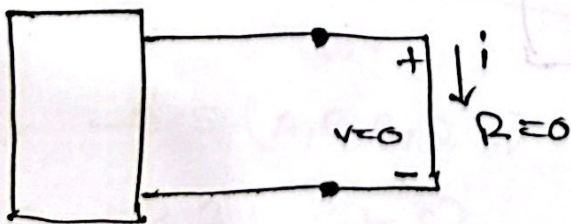
Conductor

- Silver
- Copper
- Aluminium
- Gold

* Current flows from lower to higher potential

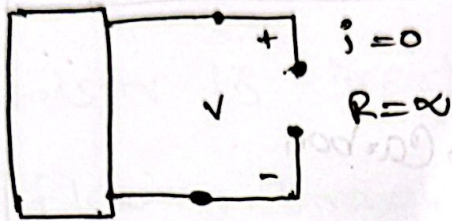
* Short circuit : ($R=0$)

A short circuit is a circuit element with resistance approaching zero, $V = iR = 0$



* Open Circuit :- ($R = \infty$)

An open circuit is a circuit element with resistance approaching infinity.



* Conductance :- (G) mhos (Ω) / siemen (S)

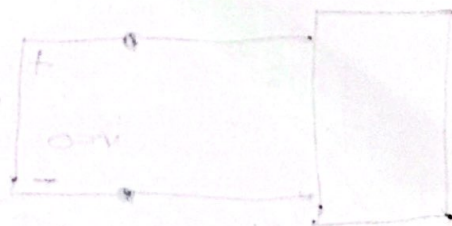
Conductance is the ability of an element to conduct electric current.

$$G = \frac{1}{R} = \frac{i}{V}$$

$$1 \text{ S} = 1 \Omega^{-1} = 1 \text{ A/V}$$

$$P = Vi = i^2 R = \frac{V^2}{R}$$

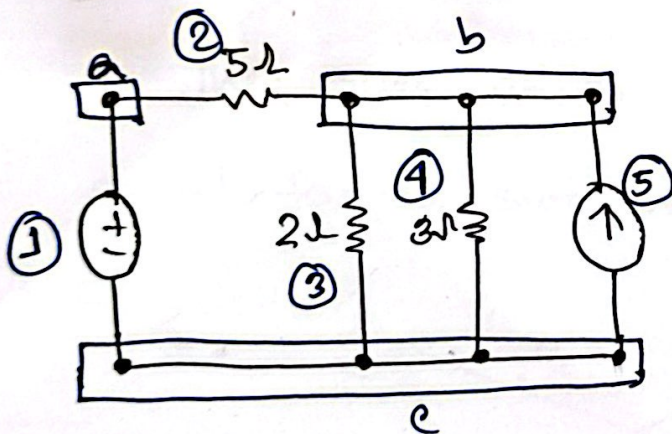
$$P = Vi = V^2 G = \frac{i^2}{G}$$



* Branch $\longrightarrow$ represents a single element in the circuit such as resistor, a voltage source.

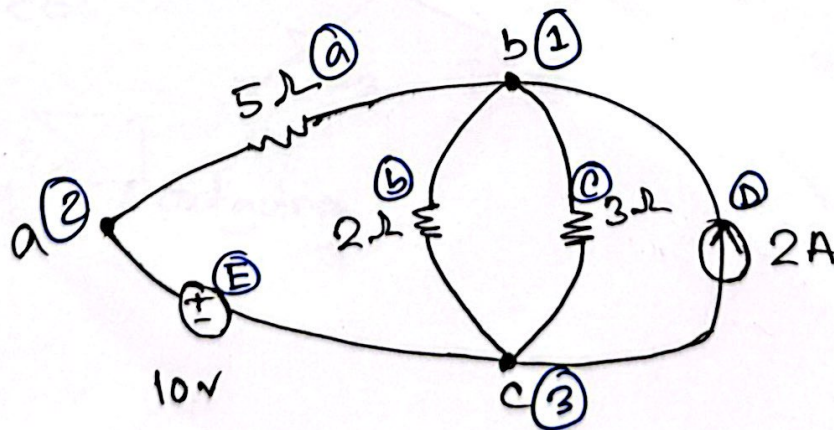
* Node $\longrightarrow$ point of connection between two or more branch

* If there is a short circuit ^(a connecting wire), connects two nodes, count as one node only.



Branch $\longrightarrow$ 5

Node $\longrightarrow$ a, b, c



Branch $\longrightarrow$ 5 (A, B, C, D, E)

Node $\longrightarrow$ 3 (a, b, c)

* Loop $\longrightarrow$ any closed path in a circuit

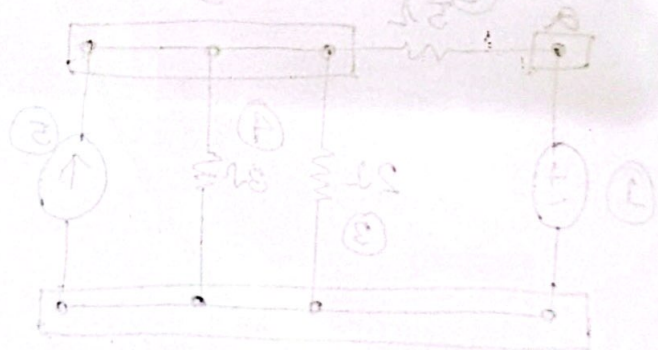
$\xrightarrow{\text{Independent loops}}$

$b = l + n - 1$

$\swarrow$ Branches $\searrow$ Nodes

* Series circuit $\longrightarrow$ Same Current

* Parallel circuit $\longrightarrow$ Same Voltage



Kirchhoff's Law

(node/junction)

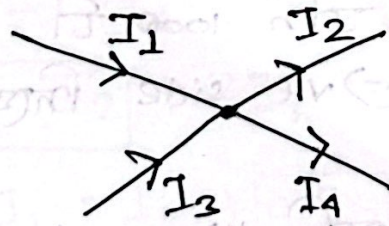
Kirchhoff's current law (KCL) → states that

the algebraic sum of currents entering a node (or a closed boundary) is zero.

$$\sum_{n=1}^N i_n = 0$$

* The sum of the currents entering a node is equal to the sum of the currents leaving the node.

$$\sum I_{\text{net}} = 0$$



$$I_1 + I_3 = I_2 + I_4$$

$$\sum I_{\text{incoming}} = \sum I_{\text{outgoing}}$$

kirchoff's voltage law :- (KVL) $\rightarrow$ (closed circuit/region)

$$\sum_{m=1}^n V_m = 0$$

$$\sum E_{net} = 0 \quad \sum V_{net} = 0$$

$$\sum E = \sum IR$$

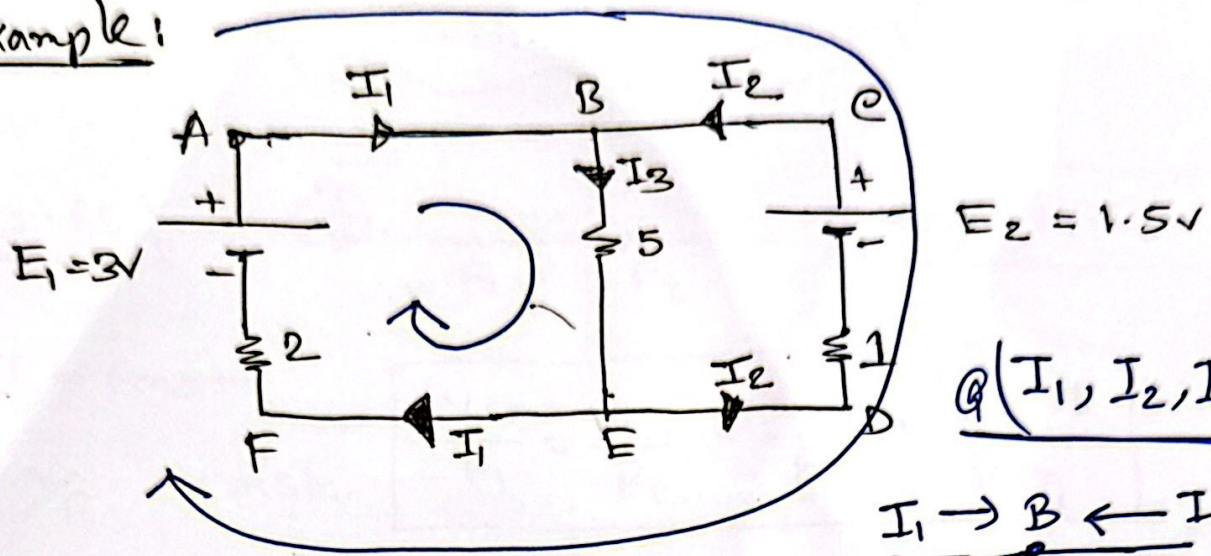
Voltage produce = Voltage dropped/lost

* KVL states that the algebraic sum of all voltages around a closed path (or loop) is zero

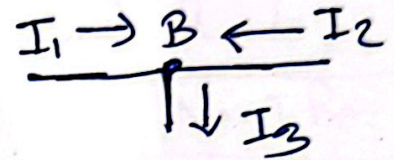
* IR $\rightarrow$ (+) if the direction of loop and current same
 $\rightarrow$ (-) If the direction is not opposite

* E $\rightarrow$ (+) $\rightarrow$ यदि loop में Battery का +ve side मिले हुए,
 $\rightarrow$ (-) यदि (+) ve side मिले हुए,

Example:



Q. ($I_1, I_2, I_3 = ?$)



Solve:

At node B using KCL, $I_1 + I_2 = I_3$

$$\Rightarrow I_1 + I_2 - I_3 = 0 \quad \text{--- (1)}$$

At loop, ABFEFA, $5I_3 + 2I_1 = 3$

$$\Rightarrow 2I_1 + \boxed{0I_2} + 5I_3 = 3 \quad \text{--- (II)}$$

For convenience of calculation.

At loop ABCDEFA,

$$\boxed{-I_2} + 2I_1 = \boxed{-1.5} + 3$$

Loop goes opp. direction of I_2

pos. side into loop I_2

$$\Rightarrow 2I_1 - I_2 + \boxed{0I_3} = 1.5 \quad \text{--- (III)}$$

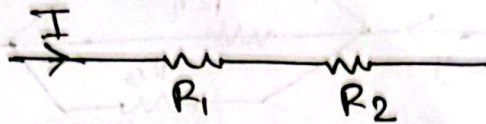
for convenience of cal.

Solving, eq (1), (II) and (III) $\rightarrow I_1 = () \quad I_2 = ()$

$$I_3 = () \quad \text{(Ans.)}$$

Combination of circuits

Series:



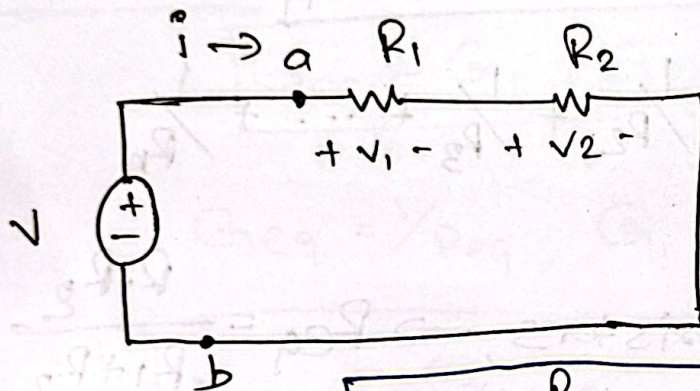
$$R_{eq} = R_1 + R_2 + \dots + R_N$$

$$I = \text{const.}$$

$$\Rightarrow \frac{V}{R} = \text{const.}$$

$$\frac{V_1}{R_1} = \frac{V_2}{R_2} = \frac{V_3}{R_3} = \frac{V_{eq} = E}{R_{eq}}$$

$$V = I R_{eq}$$



$$V_1 = \frac{R_1}{R_1 + R_2} V$$

$$V_2 = \frac{R_2}{R_1 + R_2} V$$

$$V_n = \frac{R_n}{R_1 + R_2 + \dots + R_N} V$$

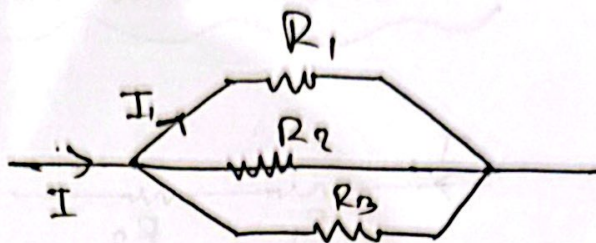
voltage divider
Law

* parallel :-

$$V = \text{Const}$$

$$\Rightarrow IR = \text{Const}$$

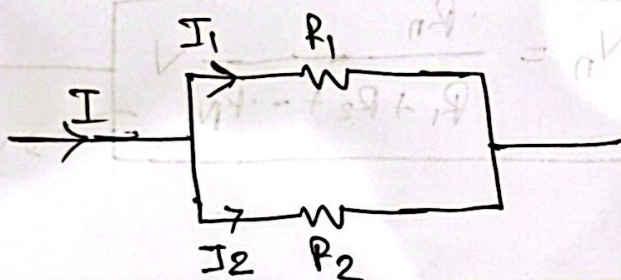
$$\Rightarrow I_1 R_1 = I_2 R_2 = I_3 R_3 = I_{\text{total}} R_{\text{eq}}$$



$$I_{\text{total}} = \frac{E}{R_{\text{eq}}}$$

$$R_{\text{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

In case of two resistors, $\rightarrow R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$



$$I_1 = \frac{R_2}{R_1 + R_2} I, \quad I_2 = \frac{R_1}{R_1 + R_2} I$$

Current div. law

* For multiple branches,

$$I_n = \frac{\frac{1}{R_n}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_n} + \frac{1}{R_{n+1}}} \times I$$

Eq. conductances:

Parall The eq. conductance of resistors connected in parallel is the sum of their individual conductances.

Parall :

$$G_{eq} = G_1 + G_2 + \dots + G_N$$

$$G_{eq} = 1/R_{eq}, G_1 = 1/R_1, G_2 = 1/R_2 \dots$$

$$i_n = \frac{G_n}{G_1 + G_2 + \dots + G_N} i_{total}$$

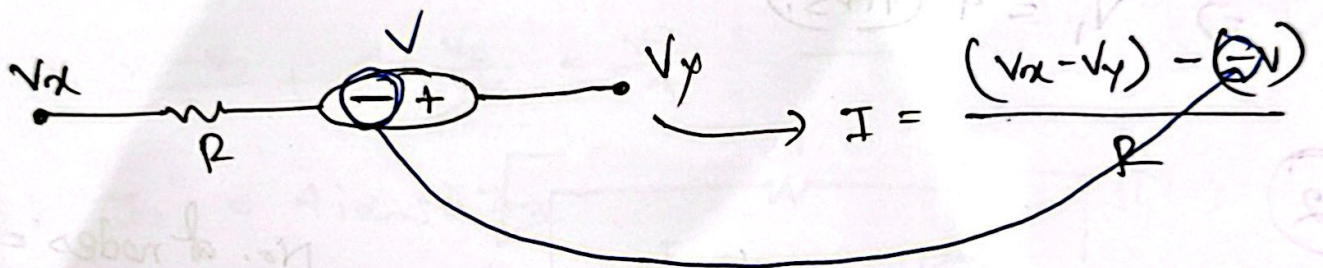
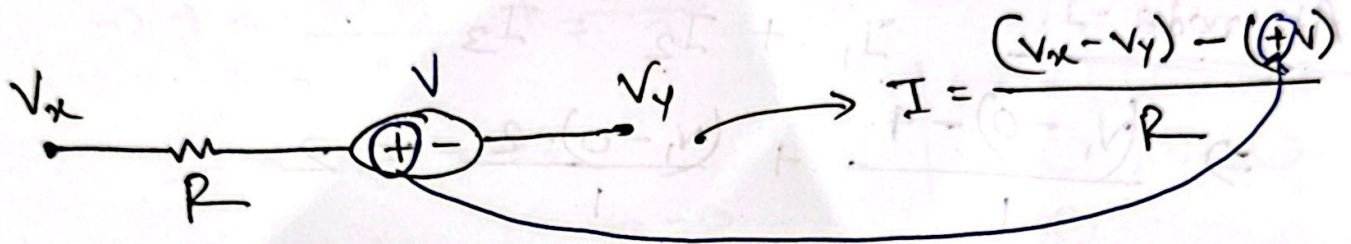
series :

$$\frac{1}{G_{eq}} = \frac{1}{G_1} + \frac{1}{G_2} + \dots + \frac{1}{G_N}$$

Nodal Analysis (KCL)

Steps:-

- (i) Specify Nodes
- (ii) select a Ref. node ($V=0$)
- (iii) Develop the KCL eq. at nodes
- (iv) solve the KCL eqs.



* Voltage source থাকলে ঐ sign নিয়ে current দু'করে গুণাই formula গ বসাবে।

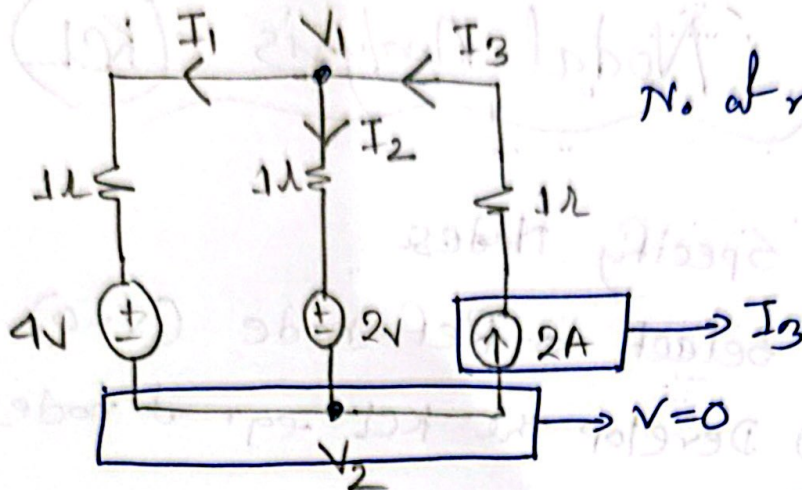
এখানে $V_x \rightarrow V_y$ arrow দেখা হয়েছে তাই বলে

V_x (বা V_y) current তে চাচ্ছে that means

$V_x > V_y$.

Example

#1



No. of nodes = 2

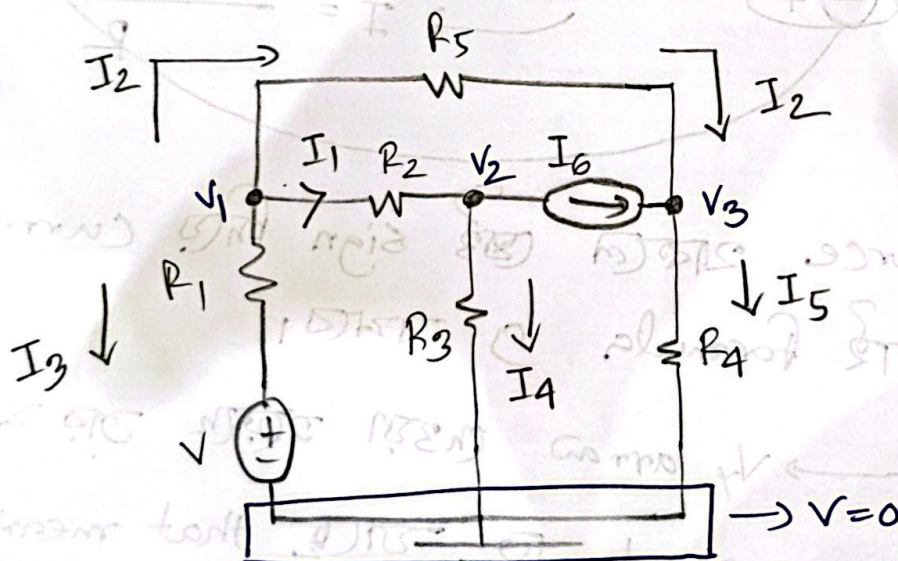
At node -1:

$$I_1 + I_2 = I_3$$

$$\Rightarrow \frac{(V_1 - 0) - 4}{1} + \frac{(V_1 - 0) - 2}{1} = 2$$

$$\Rightarrow V_1 = 4 \text{ (Ans.)}$$

#2



No. of nodes = 4

N-1: $I_1 + I_2 + I_3 = 0$

$$\Rightarrow \frac{V_1 - V_2}{R_2} + \frac{V_1 - V_3}{R_5} + \frac{(V_1 - 0) - V}{R_1} = 0$$

$$V_1 = ()$$

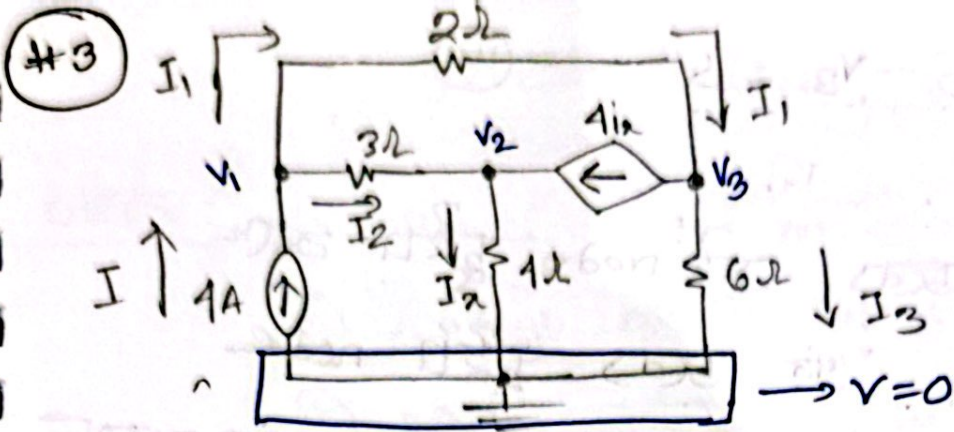
$$V_2 = ()$$

$$V_3 = ()$$

N-2: $I_6 + I_4 = I_1 \Rightarrow I + \frac{V_2 - 0}{R_3} = \frac{V_1 - V_2}{R_2}$

N-3: $I_6 + I_2 = I_5 \Rightarrow I + \frac{V_1 - V_3}{R_5} = \frac{V_3 - 0}{R_4}$

(Ans.)



N-1: $I = I_1 + I_2$

$$\Rightarrow 4 = \frac{v_1 - v_3}{2} + \frac{v_1 - v_2}{3} \quad \text{--- (i)}$$

N-2: $I_2 + 4i_x = I_x$

$$\Rightarrow \frac{v_1 - v_2}{3} + 4i_x = \frac{v_2 - 0}{1}$$

$$\Rightarrow \frac{v_1 - v_2}{3} + 4 \times \frac{v_2}{4} = \frac{v_2}{1}$$

N-3: $I_1 = 4i_x + I_3$

$$\Rightarrow \frac{v_1 - v_3}{2} = 4 \times \frac{v_2}{4} + \frac{v_3 - 0}{6} \quad \text{--- (ii)}$$

save (i), (ii), (iii)

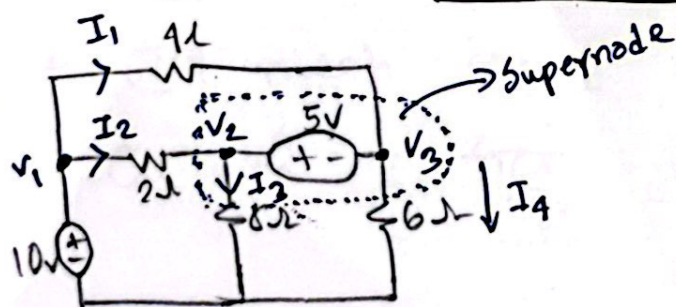
$$v_1 = 0, v_2 = 0$$

$$v_3 = 0$$

$$I_x = \left(\frac{v_2}{4} \right) = 0$$

Supernode

When a voltage source come in between two nodes.



N-1: $I_1 + I_2 = 0 \Rightarrow \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2} = 0 \quad \text{--- (i)}$

N-2, 3: (Supernode)

$$I_2 + I_1 = I_3 + I_4$$

$$\Rightarrow \frac{v_1 - v_2}{2} + \frac{v_1 - v_3}{4} = \frac{v_2}{8} + \frac{v_3}{6}$$

$$\Rightarrow () v_1 + () v_2 = () v_3 \quad \text{--- (ii)}$$

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super node $\rightarrow V_2 - V_3 = 5 \text{ — (iii)}$

* Supernode প্র ক্ষেত্রে node দুইটি আছে
ডায়েরি KCL eq. প্র ক্ষেত্রে দুইটি node
প্র incoming and outgoing প্র আছে
নিচের হবে = eq (ii) প্র মতো।

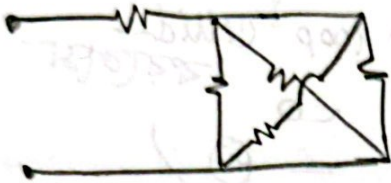
* সুধীরা নতুন Supernode প্র ক্ষেত্রে দুইটি node
প্র মধ্যকার voltage diff নিয়ে প্রকটি eq
হবে। eq (iii)

Solve করে $V_1 = ()$ $V_2 = ()$ $V_3 = ()$ (Ans)

Mesh Analysis (KVL)

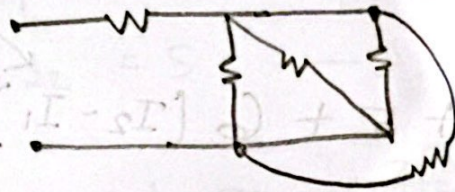
Steps:

- (i) Identify total no. of meshes
- (ii) Assign mesh currents
- (iii) Define KVL eq. for each mesh
- (iv) solve KVL eq.



Non-planar

We have to make
circuit, planar

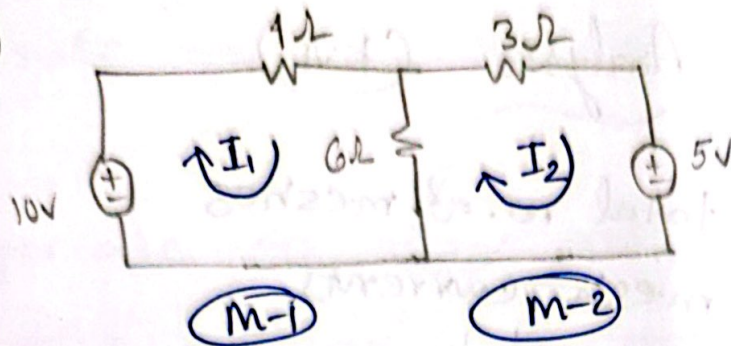


* Direction of mesh current $\rightarrow$ clockwise $\vee$
 $\rightarrow$ Anti clockwise $\times$

* If mesh current I is always greater than I , then I is greater than I .

* Resistance or sign should not be.

#1



$$I_1 \downarrow \begin{cases} 6\Omega \\ I_2(I_1 - I_2) \end{cases}$$

- * Increase of writing mesh-1 $\rightarrow I_1 > I_2$
- * Increase of writing mesh-2 $\rightarrow I_2 > I_1$

mesh-1:

$$-10 + 4I_1 + 6(I_1 - I_2) = 0 \quad \text{--- (i)}$$

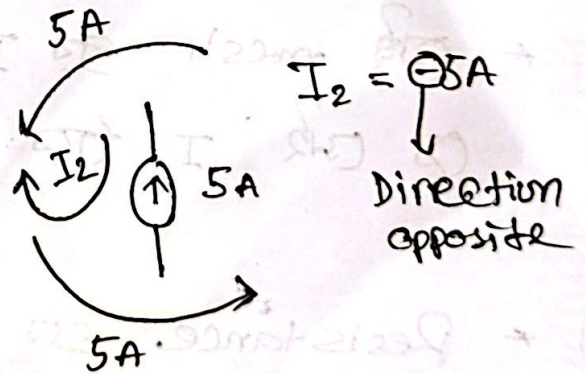
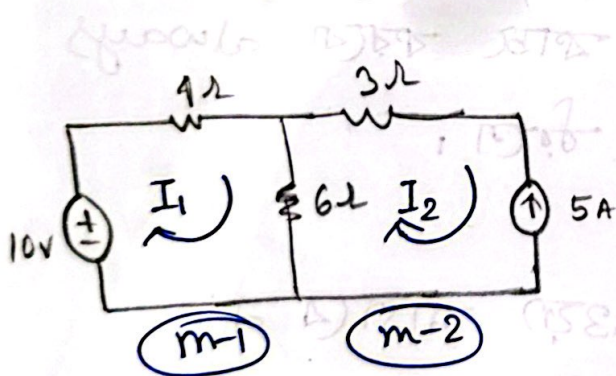
neg. sign loop collide
neg. sign loop collide

mesh-2:

$$3I_2 + 5 + 6(I_2 - I_1) = 0 \quad \text{--- (ii)}$$

Solve (i) and (ii) $\rightarrow I_1 = 0 \quad I_2 = 0 \quad \text{(Ans)}$

Mesh Analysis with current source

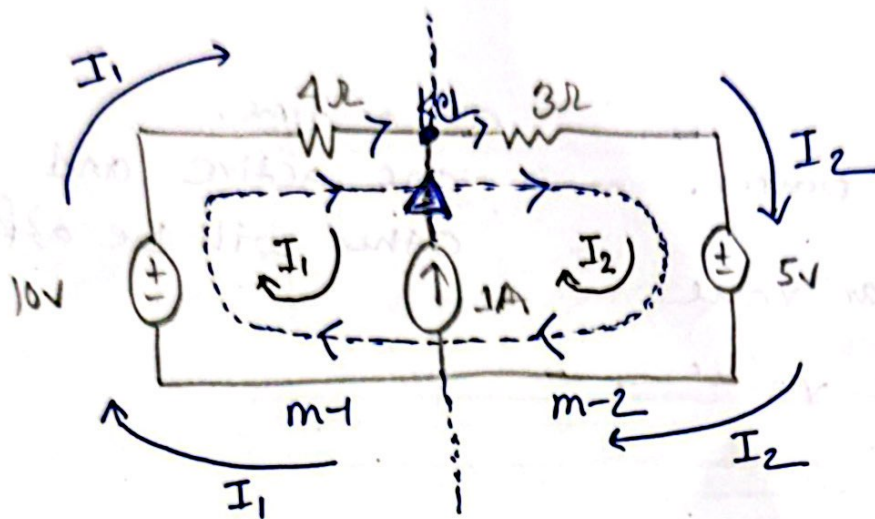


mesh-1: $-10 + 4I_1 + 6(I_1 - I_2) = 0$

$\Rightarrow I_1 = -2A$

$\rightarrow$ loop 3A assumed direction opposite

Supermesh



Supermesh $\rightarrow$
$$\underbrace{-10 + 4I_1}_{m-1} + \underbrace{3I_2 + 5}_{m-2} = 0$$

$$\Rightarrow 4I_1 + 3I_2 = 5 \quad \text{--- (1)}$$

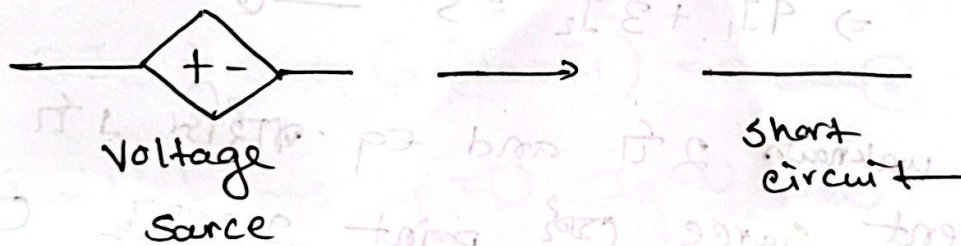
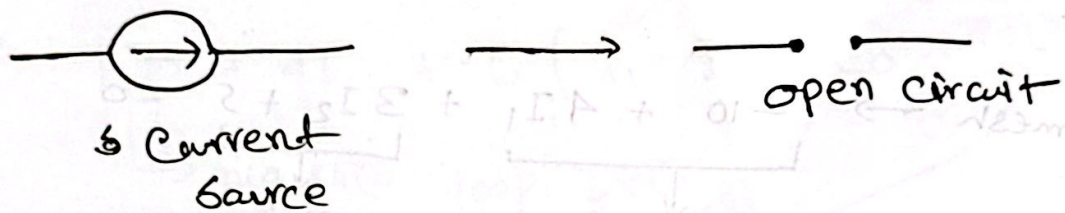
★ As unknown 2 টা and eq পাঠিয়ে ১ টা ৬০ current source ফেরে point এ যাচ্ছে ডোই Node এ KCL দিয়ে eq আবেগটা নিতে হবে।

$$I_1 + 1 = I_2 \quad \text{--- (2)}$$

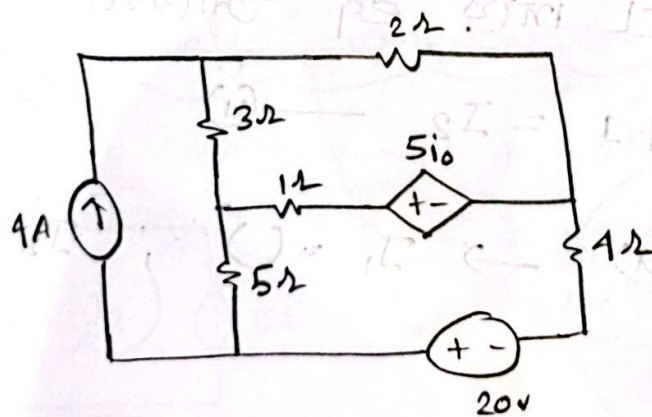
Solve (1) and (2) $\rightarrow I_1 = ()$; $I_2 = ()$ (Ans.)

Super-position

- at a time,
- (i) Independent source $\rightarrow$ one active and other will be off
 - (ii) Get particular value
 - (iii) Sum up the value

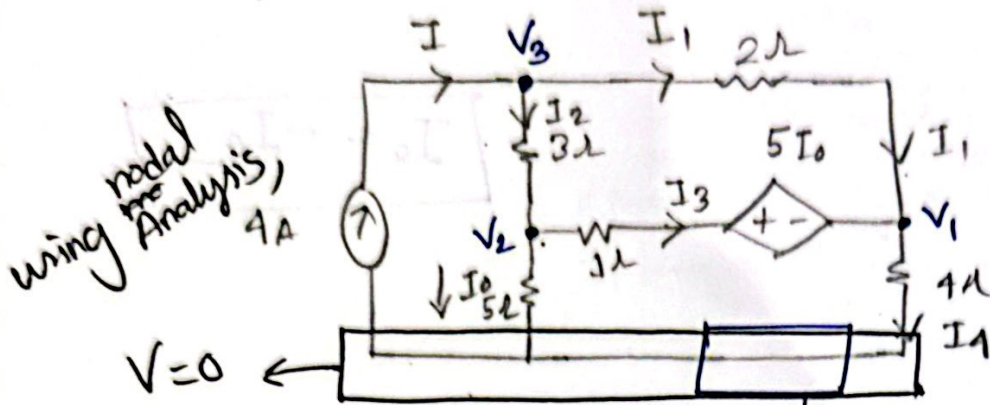


Example:



$\rightarrow$ solve using Super-position theorem.

* keeping 4A active and rest of them inactive,



N-1: $I_1 + I_3 = I_4$

$$\Rightarrow \frac{V_3 - V_1}{2} + \frac{(V_2 - V_1) - 5I_0}{1} = \frac{V_1}{4}$$

short circuit
2A 2(2/2)

N-2: $I_2 = I_0 + I_3$

$$\Rightarrow \frac{V_3 - V_2}{3} = \frac{V_2}{5} + \frac{(V_2 - V_1) - 5I_0}{1}$$

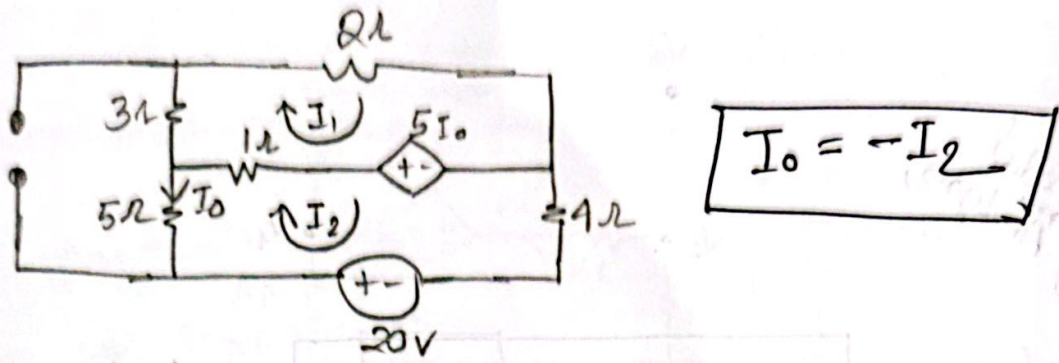
$$I_0 = \frac{V_2}{5}$$

N-3: $4 = I_1 + I_2$

$$\Rightarrow 4 = \frac{V_3 - V_1}{2} + \frac{V_3 - V_2}{3}$$

$$V_1 = () \quad V_2 = () \quad V_3 = () \quad \left[I_0' = \frac{V_2}{5} = () \right]$$

* keeping 20 V active, and rest of them inactive,



using mesh analysis,

$$\underline{m-1}: 3I_1 + 2I_1 - 5I_0 + 1(I_1 - I_2) = 0$$

$$\Rightarrow 6I_1 - I_2 - 5I_0 = 0$$

$$\underline{m-2}: -20 + 5I_2 + 1(I_2 - I_1) + 5I_0 = 0$$

$$\Rightarrow 6I_2 - I_1 + 5I_0 = 0$$

$$I_1 = () \quad I_2 = ()$$

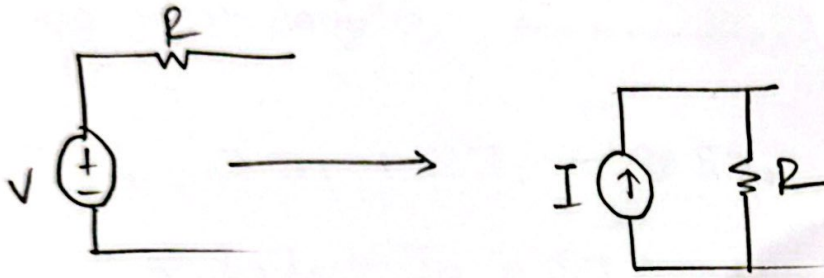
$$\boxed{I_0'' = -I_2}$$

$$\therefore \boxed{I_0 = I_0' + I_0''} \rightarrow (Ans.)$$

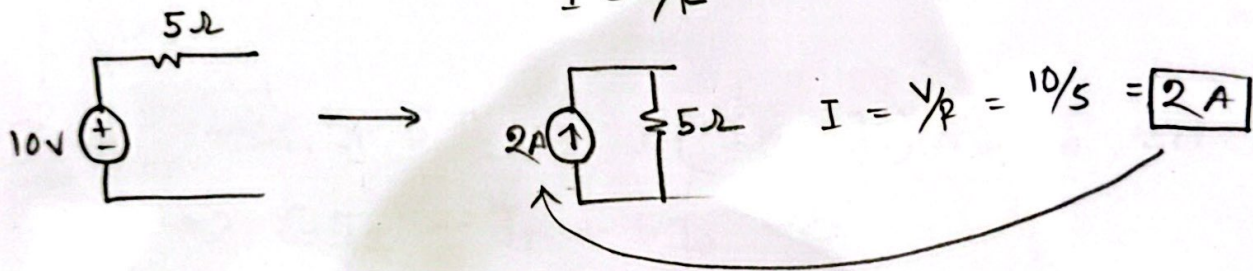
Source Transformation

① Voltage source $\longrightarrow$ Current source;

Voltage source এর সাথে series এ একটি resistance যুক্ত থাকতে হবে।

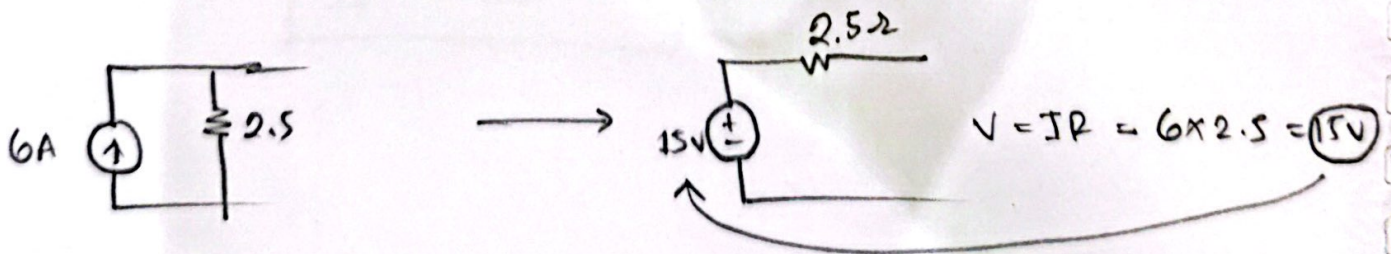


$$I = V/R$$

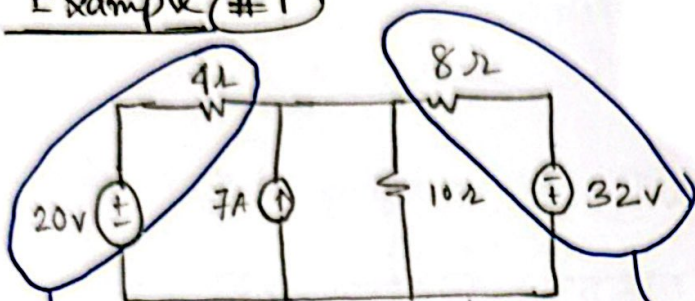


② Current source $\longrightarrow$ Voltage source:-

③ Current source এর সাথে parallel - এ একটি resistor যুক্ত থাকতে হবে।



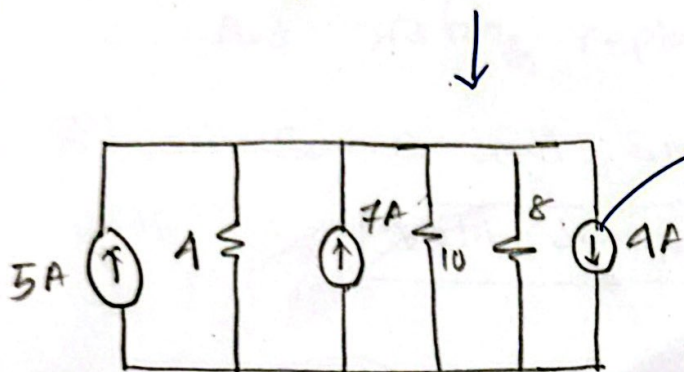
Example #1



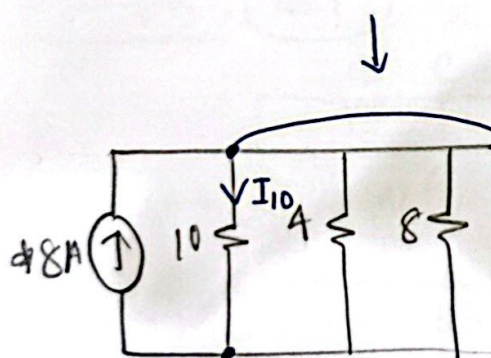
Q → Find current through the 10Ω ?

$$I_1 = \frac{20}{4} = 5A$$

$$I_2 = \frac{32}{8} = 4A$$



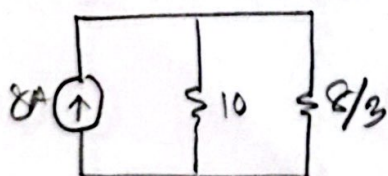
→ opposite because volt source এর polarity opposite ছিল, arrow always pos. এর নিচে থাকবে,



$$5 \uparrow \quad 7 \uparrow \quad 4 \downarrow \rightarrow [(5+7)-4]A = 8A$$

$$(10 \parallel 4 \parallel 8)$$

"10Ω" এ calc. করতে হবে
এটা তার Req কে বসানো



using current divider rule,

$$I_{10} = \frac{8/3}{10 + 8/3} \times 8 = 1.684A \quad (Ans)$$

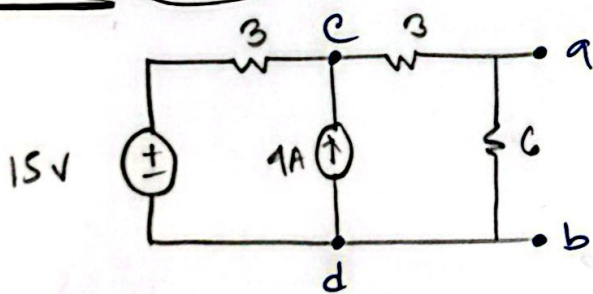
Thevenin Theorem (Independent source)

⇒ এই point এ Thevenin Theo. we করে circuit কেবল করতে বলবে যেহেঁত **LEFT** part নিয়ে কাজ করতে হবে,

And সব কিছু replace হবে Just একটি single 8 volt. source and single resistance in **Series** with **সেই Volt. source** থাকবে,

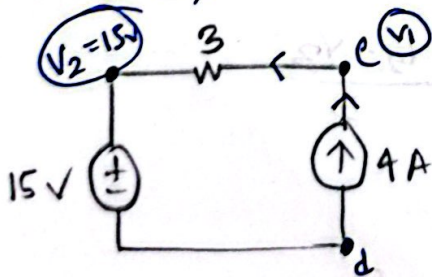
$$I_{TH} = \frac{V_{TH}}{R_{TH}}$$

Example: #1



Q → Det. Thev. eq. circuit at c-d and a-b?

For c-d,



$$V_{TH} \Rightarrow 4A = \frac{V_1 - V_2}{3} = \frac{V_1 - 15}{3}$$

$$\Rightarrow V_{TH} = 27V$$

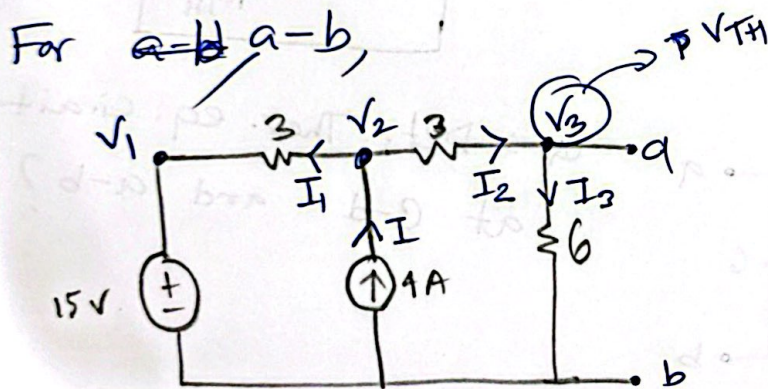
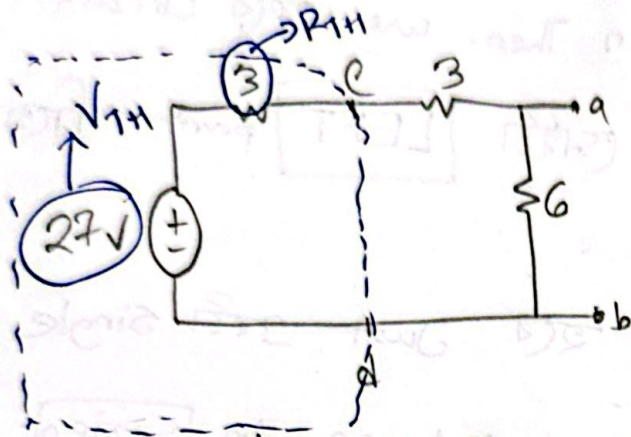
→ R_{TH} কেবল করতে অন্য এই দুই point

এর মতো শুধু V. source and current. source
তারা এইসকল সব turn off করতে হবে,

V. source → short

Current source → Open

Carfilez
Carfilzomib INN Inj.



N-1 $\rightarrow V_1 = 15V$

N-2 $\rightarrow 4 = I_1 + I_2 = \frac{V_2 - 15}{3} + \frac{V_2 - V_3}{3}$

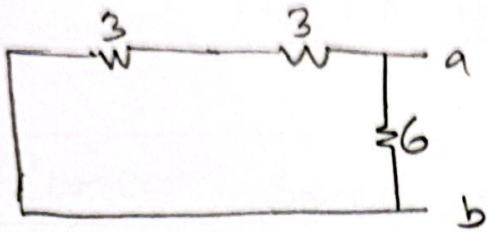
$\Rightarrow 27 = 2V_2 - V_3 \quad \text{--- (i)}$

N-3 $\rightarrow I_2 = I_3 \Rightarrow \frac{V_2 - V_3}{3} = \frac{V_3 - 0}{6}$

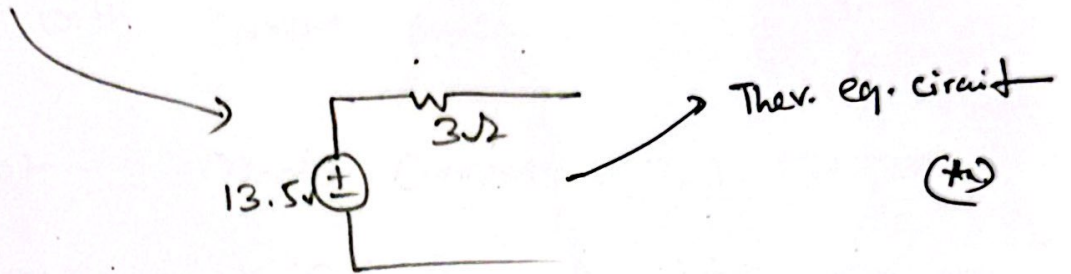
$\Rightarrow 2V_2 = 3V_3 \quad \text{--- (ii)}$

Solving (i) and (ii) $\rightarrow V_3 = 13.5V \rightarrow V_{TH} \text{ at } a$

Na₇



$$R_{TH} = (3+3) \parallel 6 = 3\Omega$$



Thev. eq. circuit
(*)

Norton Theorem

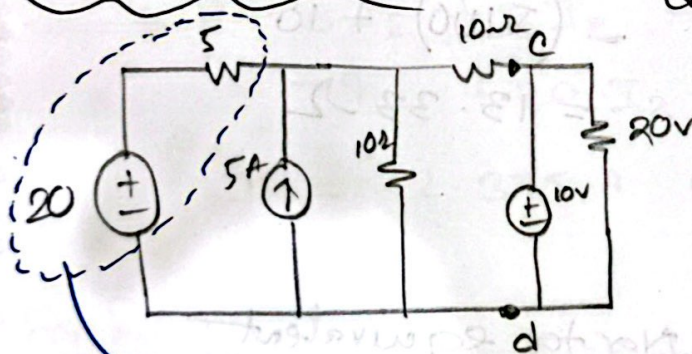
⇒ Current source and resistor parallel -
 সাথে with current source.

⇒ যেই point -এ Norton Current (I_N) বের করতে হবে সেই point এ short circuit করার পর যেই current টি চাবে তেই short circuit দিয়ে
 চিহ্নিত I_N

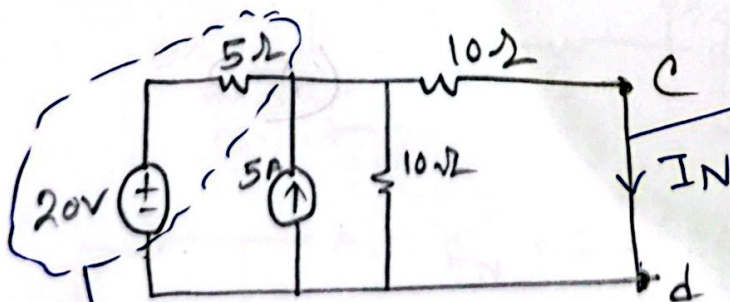
⇒ $R_N \longrightarrow$ Same as Thevenin circuit

Example : #1

Q → Determine Norton equivalent circuit at c-d?



→ same transformation করে দেওয়া হবে।

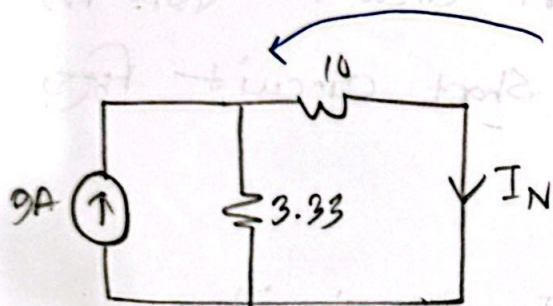
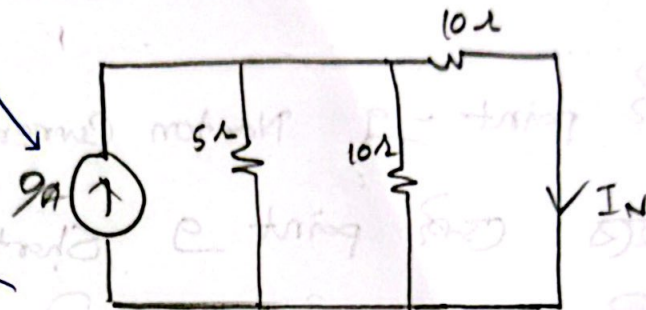
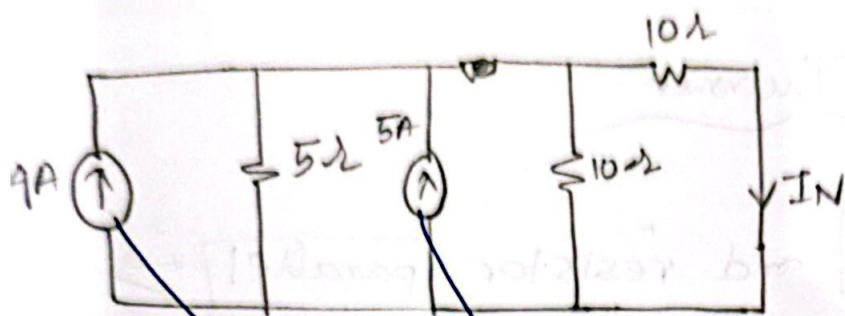


→ Short circuit

→ same transformation হবে

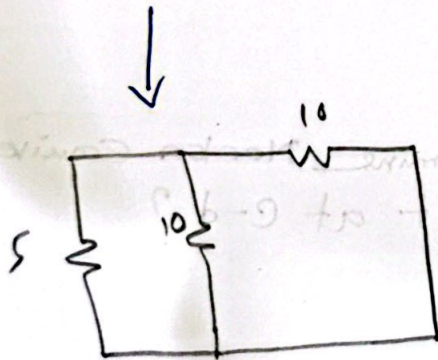
$$I = \frac{V}{R} = \frac{20}{5} = 4A$$

Carfilez
 Carfilezomib INN-Inj.

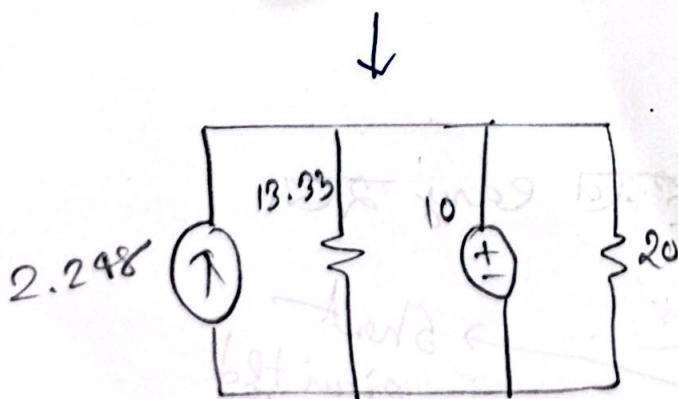


using current div. Rule,

$$I_N = \frac{3.33}{13.33} \times 9 = 2.248A$$



$$R_N = (5 \parallel 10) + 10 = 13.33 \Omega$$

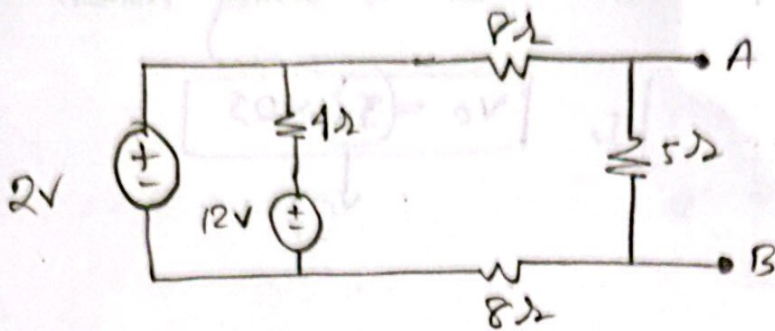


Norton equivalent circuit at a-d.

(Ans.)

#2

Def. N. eq. circuit at A-B,



As A-B point
is shorted so find
I_{sc} point

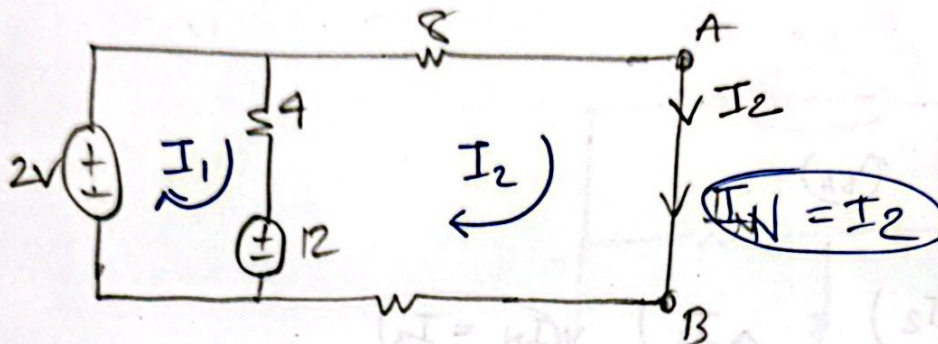
short circuit
voltage is 0

because of this
there is 5Ω resistor

total current

through AB
path is what
any resistance
current can
flow through

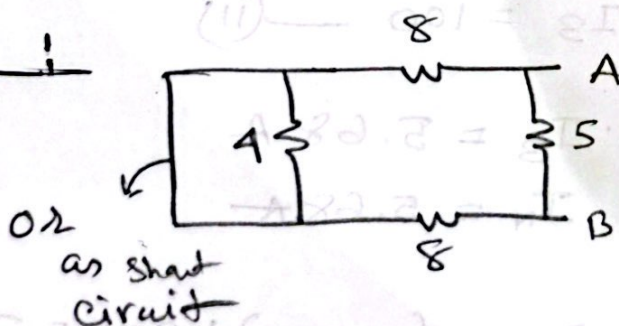
①



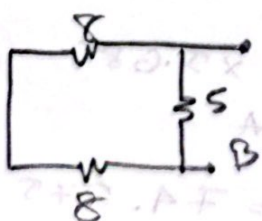
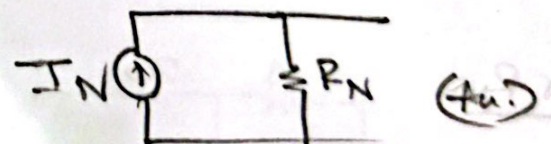
m-1: $-2 + 4(I_1 - I_2) + 12 = 0$
 $\Rightarrow 4I_1 - 4I_2 = -10$

m-2: $8I_2 + 8I_2 - 12 + 4(I_2 - I_1) = 0$
 $-4I_1 + 20I_2 = 12$

$I_1 = -2.375 \text{ A}$ $I_2 = 0.125 \text{ A} = I_N$

For R_N :

① $0.125 \text{ A} = \frac{0 \times 4}{0 + 4} = 0.125$

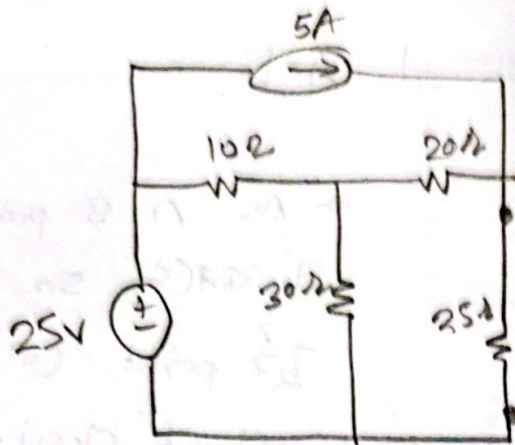


$R_N = 5 \parallel (8 + 0 + 5)$
 $= 5 \parallel 13$

$= \frac{5 \times 13}{5 + 13} = 3.8095 \Omega$

Carfilez
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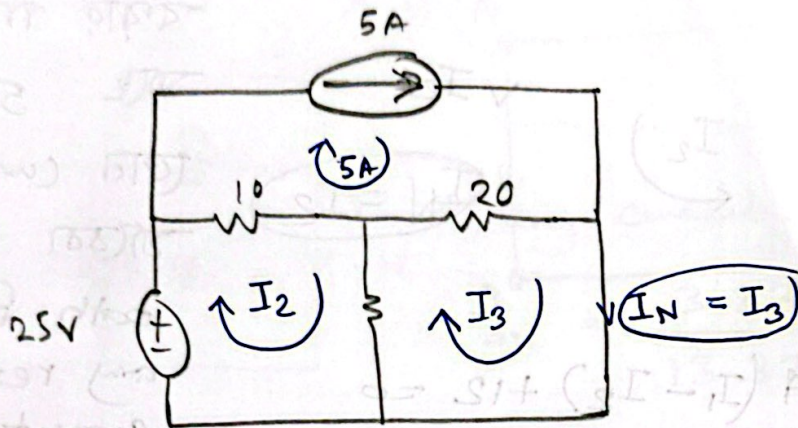
#3



Q → Find V_o using Norton

$$V_o = I_o \times 25$$

↓ ??

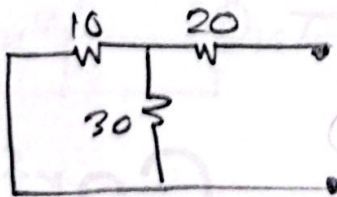


m-2: $-25 + 10(I_2 - 5) + 30(I_2 - I_3) = 0$
 $\Rightarrow 40I_2 - 30I_3 = 75$ — (i)

m-3: $20(I_3 - 5) + 30(I_3 - I_2) = 0$
 $\Rightarrow -30I_2 + 50I_3 = 100$ — (ii)

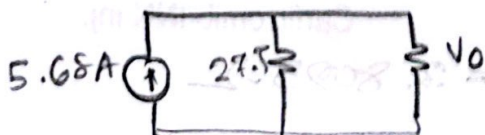
Solving (i) and (ii) $\rightarrow I_3 = 5.68 \text{ A}$
 $I_N = 5.68 \text{ A}$

For R_N :



$$R_N = (10 \parallel 30) + 20 = 27.5 \Omega$$

$$I_o = \frac{27.5}{27.5 + 25} \times 5.68 = 2.975 \text{ A}$$

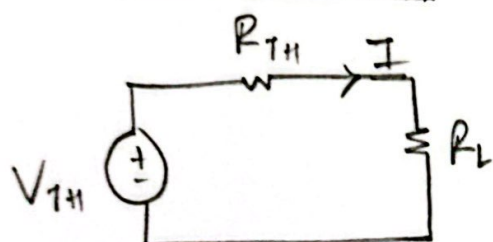


$$V_o = I_o \times 25 = 74.375 \text{ V}$$

Maximum Power Theorem

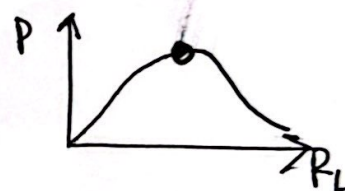
Maximum power is transferred from source to load when the load resistance equals to thevenin resistance as seen from load. $P = I^2 R_L$

$R_L = R_{TH}$ → then get maximum power at.

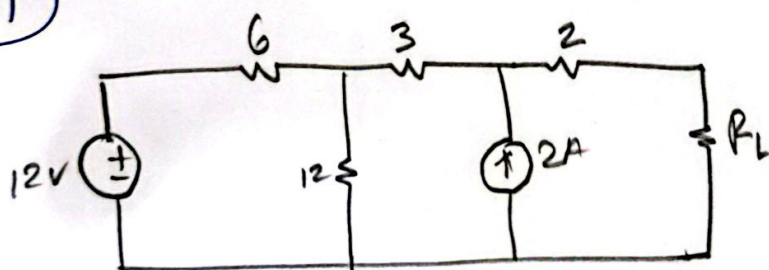


$$I \uparrow = R_L \downarrow$$

$$P_{max} = \frac{V_{TH}^2}{4 R_{TH}}$$



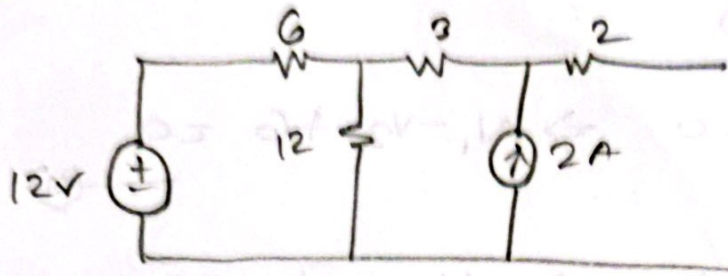
#1



Q → Find the value of R_L for maximum power transfer from the following circuit and also find the maximum power?

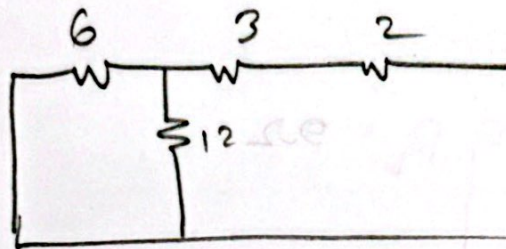
Steps:

① Remove R_L from the circuit



② Determine thevenin equivalent circuit

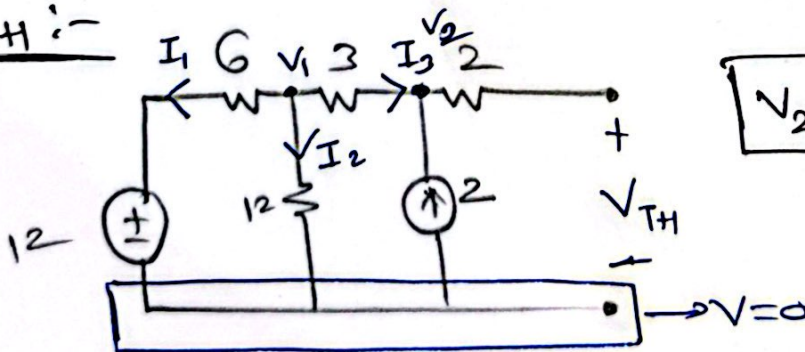
For R_{TH} :



$$R_{TH} = (6 \parallel 12) + 3 + 2 = 9\Omega$$

$$R_L = 9\Omega \quad [\text{as max power } R_{LH} = R_L]$$

For V_{TH} :



$$V_2 = V_{TH}$$

N-1: $I_1 + I_2 + I_3 = 0$

$$\frac{(V_1 - 0) - 12}{6} + \frac{V_1}{12} + \frac{V_1 - V_2}{3} = 0$$

$$\Rightarrow \frac{V_1 - 12}{6} + \frac{V_1}{12} + \frac{V_1 - V_2}{3} = 0$$

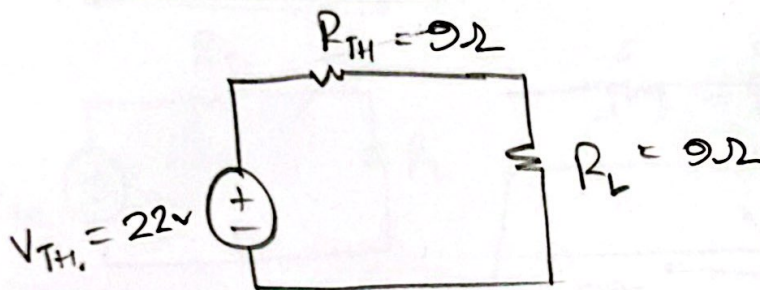
$$\Rightarrow 7V_1 - 4V_2 = 24 \quad \text{--- (1)}$$

N-2! $I_3 + 2 = 0$

$$\Rightarrow \frac{V_1 - V_2}{3} + 2 = 0 \Rightarrow V_1 - V_2 + 6 = 0 \quad \text{--- (2)}$$

Solving (1) and (2) $\Rightarrow V_1 = 16V ; V_2 = 22V$

$$V_{TH} = 22V$$



$$P_{max} = \frac{V_{TH}^2}{4 R_{TH}} = \frac{(22)^2}{4 \times 9} = 13.44 \text{ W (Ans)}$$

$$11V = 5V$$